

D664

Learners and Learning Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A five-year-old watches water poured from a short, wide glass into a tall, thin glass and says the tall glass has more. She is the youngest of four siblings and loves painting. According to Piaget, what does this show?

- A. She is behind because older siblings do things for her.
- B. She is in the concrete operational stage.
- C. She is in the preoperational stage and has not yet developed conservation.
- D. She is in the formal operational stage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A student can solve “If all zorbs are blue and this is a zorb, what color is it?” and can plan a fair test by thinking through variables not in front of her. Which stage does this reasoning suggest?

- A. Preoperational
- B. Sensorimotor
- C. Formal operational
- D. Concrete operational

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A student can solve two-step word problems when the teacher asks guiding questions, but not on his own. He sits near the window and is the class line leader this week. According to Vygotsky, where should instruction focus?

- A. On two-step problems with guided support that is gradually removed.
- B. Move his seat away from the window first.
- C. On one-step problems he can already do alone.
- D. On multistep algebra problems with no help.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A student now solves two-step problems correctly without any prompts. What should happen to the scaffolding?

- A.** Remove it and move to a new challenge in the next ZPD.
- B.** Return to easier problems.
- C.** Keep all the prompts in place permanently.
- D.** Add more prompts to be safe.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A teacher explains a new 9-step science procedure all at once, then asks students to begin. The lab has new stools and a large periodic table poster. Most students forget steps. What is the best change?

- A.** Take down the poster to reduce distractions.
- B.** Explain all 9 steps again, faster.
- C.** Add more details to each step so it is complete.
- D.** Break the procedure into small chunks, model each, and provide a step card.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Students are preparing for a test on the causes of the American Revolution. The class uses a colorful textbook and studies in groups of four. Which review strategy is best supported by learning science?

- A.** Reread the chapter twice.
- B.** Close the book and answer practice questions from memory, then check answers.
- C.** Highlight key sentences in the textbook.
- D.** Rely on the textbook's colorful images to remember.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A teacher has 60 minutes for vocabulary review before a test two weeks away. She teaches in a portable classroom and has a new smartboard. Which schedule should produce the best long-term retention?

- A. Six 10-minute sessions spread across the two weeks.
- B. One 60-minute session the day before the test.
- C. One 60-minute session today, then none.
- D. It does not matter, as long as the smartboard is used.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A student crams for 4 hours the night before a test and scores well, but a month later remembers little. Which principle explains this?

- A. The student is a poor learner.
- B. The student needed to highlight more.
- C. Massed practice supports short-term performance but not long-term retention; spacing would help.
- D. Cramming always produces permanent learning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A teacher tells students that anyone who finishes classwork with care will not have homework that night. Work completion goes up. The class meets in room 214, and many students ride the bus. What is this an example of?

- A. Extinction
- B. Negative reinforcement
- C. Positive reinforcement
- D. Punishment

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A teacher wants to decrease students calling out answers. Which consequence is a form of punishment?

- A.** Homework is canceled for students who call out.
- B.** Any consequence is punishment.
- C.** Students who raise hands earn a sticker.
- D.** A student who calls out loses one minute of free choice time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher gives a learning-styles survey and plans to teach “visual learners” only with pictures and “auditory learners” only with lectures. The survey was free online and took 10 minutes. What does research suggest?

- A.** Use a longer survey for more accurate labels.
- B.** The plan is sound because every student has one fixed learning style.
- C.** Teach every lesson only through lecture.
- D.** Matching instruction to learning-style labels is not supported; choose modes that fit the content for everyone.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Students say they studied and felt ready, but most scored poorly on a quiz. The quiz was on a Monday after a long weekend. Which strategy best targets the underlying issue?

- A.** Tell students to study longer the same way.
- B.** Teach students to self-test and rate their confidence before checking answers, then compare.
- C.** Assume students always know when they understand.
- D.** Never give quizzes on Mondays.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A five-year-old watches water poured from a short, wide glass into a tall, thin glass and says the tall glass has more. ~~She is the youngest of four siblings and loves painting.~~ According to Piaget, what does this show?

Target word: Piaget stage **Decides:** Lacks conservation.

A	Camouflage	Family details do not explain a typical stage feature.
B	Adjacent category	Concrete operational children have conservation.
C	Correct	Lack of conservation is typical of preoperational thinking.
D	Confusion set	Formal operations bring abstract reasoning, far beyond this.

2. Answer C Concept 1 · transfer

A student can solve “If all zorbs are blue and this is a zorb, what color is it?” and can plan a fair test by thinking through variables not in front of her. Which stage does this reasoning suggest?

Target word: reverse: stage from abstract reasoning **Decides:** Hypothetical, abstract reasoning.

A	Inverted operation	Preoperational thinking is not yet logical.
B	Confusion set	Infancy stage based on senses and actions.
C	Correct	Abstract and hypothetical reasoning marks formal operations.
D	Adjacent category	Logic is tied to concrete objects at that stage.

3. Answer A Concept 2

A student can solve two-step word problems when the teacher asks guiding questions, but not on his own. ~~He sits near the window and is the class line leader this week.~~ According to Vygotsky, where should instruction focus?

Target word: ZPD **Decides:** Succeeds with help, not alone.

A	Correct	This is his ZPD; scaffold then fade.
B	Camouflage	Seating does not locate the ZPD.
C	Inverted operation	Below the ZPD; little new learning.
D	Adjacent category	Beyond the ZPD.

4. Answer A Concept 2 · transfer

A student now solves two-step problems correctly without any prompts. What should happen to the scaffolding?

Target word: reverse: fading support **Decides:** Competence is independent now.

A	Correct	Scaffolds fade as competence grows.
B	Dropped element	Drops the goal of moving learning forward.
C	Absolute language	Permanent scaffolds block independence.
D	Inverted operation	Increases support when less is needed.

5. Answer D Concept 3

A teacher explains a new 9-step science procedure all at once, then asks students to begin. ~~The lab has new stools and a large periodic table poster.~~ Most students forget steps. What is the best change?

Target word: reduce cognitive load **Decides:** Too many new steps at once for working memory.

A	Camouflage	The poster is not the main source of overload.
B	Inverted operation	Speed adds load.
C	Adjacent category	Extra detail raises extraneous load.
D	Correct	Chunking and external supports reduce working memory load.

6. Answer B Concept 4

Students are preparing for a test on the causes of the American Revolution. ~~The class uses a colorful textbook and studies in groups of four.~~ Which review strategy is best supported by learning science?

Target word: effective study strategy **Decides:** Retrieval strengthens memory.

A	Confusion set	Rereading feels fluent but builds less retention.
B	Correct	Retrieval practice strengthens long-term learning.
C	Adjacent category	Highlighting is a low-utility strategy.
D	Camouflage	Design features do not replace retrieval.

7. Answer A Concept 5

A teacher has 60 minutes for vocabulary review before a test two weeks away. ~~She teaches in a portable classroom and has a new smartboard.~~ Which schedule should produce the best long-term retention?

Target word: spacing **Decides:** Distributed practice beats massed practice.

A	Correct	Spacing improves long-term retention.
B	Inverted operation	Massed cramming fades quickly.
C	Dropped element	No later practice to strengthen memory.
D	Camouflage	The tool does not decide retention.

8. Answer C Concept 5 • transfer

A student crams for 4 hours the night before a test and scores well, but a month later remembers little. Which principle explains this?

Target word: reverse: explain the outcome **Decides:** Massed practice helps short-term, not long-term.

A	Adjacent category	Blames the learner instead of the strategy.
B	Confusion set	Another low-utility strategy.
C	Correct	The spacing effect.
D	Absolute language	Contradicted by the outcome.

9. Answer B Concept 6

A teacher tells students that anyone who finishes classwork with care will not have homework that night. Work completion goes up. ~~The class meets in room 214, and many students ride the bus.~~ What is this an example of?

Target word: operant conditioning term **Decides:** An unpleasant thing is removed and behavior increases.

A	Inverted operation	Extinction withholds reinforcement so behavior decreases.
B	Correct	Removing homework increases careful work.
C	Adjacent category	Positive reinforcement adds something pleasant.
D	Confusion set	Punishment decreases a behavior.

10. Answer D Concept 6 • transfer

A teacher wants to decrease students calling out answers. Which consequence is a form of punishment?

Target word: reverse: pick the punishment **Decides:** Punishment decreases behavior.

A	Inverted operation	Removing something unpleasant would increase calling out.
B	Absolute language	Consequences can reinforce or punish.
C	Adjacent category	Positive reinforcement of a different behavior.
D	Correct	Removing something pleasant to decrease behavior is negative punishment.

11. Answer D Concept 7

A teacher gives a learning-styles survey and plans to teach “visual learners” only with pictures and “auditory learners” only with lectures. ~~The survey was free online and took 10 minutes.~~ What does research suggest?

Target word: learning styles neuromyth **Decides:** Matching to style labels does not improve learning.

A	Camouflage	The survey length is not the issue.
B	Absolute language	Preferences exist, but teaching to them does not improve outcomes.
C	Inverted operation	Rejecting styles does not mean using one mode.
D	Correct	Research does not support the meshing hypothesis.

12. Answer B Concept 8

Students say they studied and felt ready, but most scored poorly on a quiz. ~~The quiz was on a Monday after a long weekend.~~ Which strategy best targets the underlying issue?

Target word: metacognition **Decides:** Students misjudged their own understanding.

A	Dropped element	Drops the monitoring problem.
B	Correct	Builds monitoring of their own learning.
C	Absolute language	The evidence shows they did not.
D	Camouflage	Reacts to timing, not the misjudgment.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Piaget's stages	Preoperational children (about 2–7) lack conservation; concrete operational children (about 7–11) reason logically about concrete objects; formal operational (about 11+) reason abstractly and hypothetically.	Mixing up which stage brings conservation vs. abstract reasoning.	A child's reasoning about quantity or a hypothetical.
2	Zone of proximal development and scaffolding	Instruction is most effective in the ZPD—what a learner can do with help but not yet alone—with support faded as competence grows.	Teaching only what students can already do alone, or never removing support.	A student who succeeds with prompts but not independently.
3	Working memory and cognitive load	Working memory holds only a few new items at once; reduce extraneous load and chunk new material.	Presenting many new steps at once to save time.	Students lost during a long multistep explanation.
4	Retrieval practice	Recalling information from memory strengthens learning more than rereading it.	Believing rereading and highlighting are the best ways to study.	Choosing a review strategy.
5	Spaced practice	Spreading practice over time produces better long-term retention than massing it in one session.	Cramming because it feels effective.	Scheduling review across days or weeks.
6	Reinforcement vs. punishment	Reinforcement (positive or negative) increases a behavior; punishment decreases it. Negative reinforcement removes something unpleasant.	Calling negative reinforcement a punishment.	Something is removed and a behavior goes up.
7	Neuromyths: learning styles	Matching instruction to a student's preferred learning style has not been shown to improve learning; match the mode to the content.	Labeling students as visual, auditory, or kinesthetic learners and teaching only that way.	A plan built around learning-style labels.
8	Metacognition	Learners improve when they plan, monitor, and evaluate their own thinking and learning.	Assuming students naturally know when they do not understand.	Students who think they know material but perform poorly.

Course scope (public description): Learners and Learning Science provides WGU students with a deep understanding of the science behind learning processes. This course covers a broad spectrum of topics pertaining to the science of learning, including cognitive development, learning theories, neuroscience in education, and the impact of developmental milestones on learning. Students will explore how these concepts apply to learning environments and educational levels, from early childhood through adolescence. The course emphasizes evidence-based practices and the practical application of learning science principles, equipping students with strategies to enhance learning outcomes and student engagement. This course aims to empower educators to create more effective, inclusive, and engaging learning experiences for all learners.